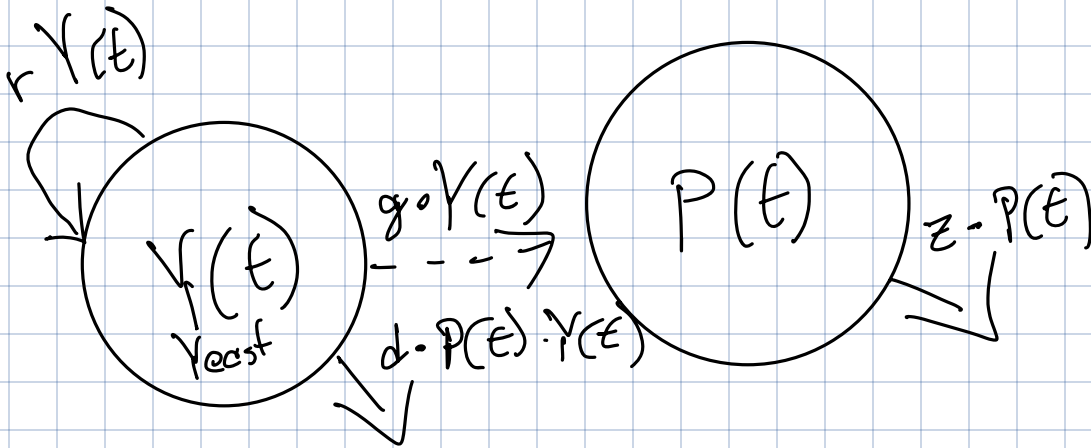


$$0 < g \leq 1$$

$$0 < d \leq 1$$

$$0 < z \leq 1$$

$$0 < r \leq 1$$



$$\frac{dY}{dt} = rY - dPY$$

$$\frac{dP}{dt} = gY - zP$$

$$g\hat{Y} - z\hat{P} = 0$$

$$\hat{P} = \frac{g}{z}\hat{Y}$$

$$0 = rY - dPY \rightarrow \boxed{\hat{Y} = 0} \rightarrow \boxed{\hat{P} = 0}$$

$$0 = r - dP$$

$$0 = r - \frac{d}{g}Y$$

$$\boxed{\hat{Y} = \frac{rz}{d}}$$

$$\rightarrow \boxed{\hat{P} = \frac{r}{d}}$$

Jacobian Matrix

$$J = \begin{pmatrix} \frac{dY}{dY} & \frac{dY}{dP} \\ \frac{dP}{dY} & \frac{dP}{dP} \end{pmatrix} =$$

$$\begin{pmatrix} r - dP & -dY \\ g & -z \end{pmatrix}$$

$$\begin{pmatrix} 0 & -\frac{rz}{d} \\ 0 & -z \end{pmatrix}$$

$$J \begin{vmatrix} \hat{p} = \frac{r}{d}, & \hat{y} = \frac{rz}{dg} \\ & \end{vmatrix} = \begin{pmatrix} 0 & g \\ d & -z \end{pmatrix}$$

$$\lambda_1 = \frac{-gz - \sqrt{g^2 z(z-4r)}}{2g} \quad \lambda_2 = \frac{-gz + \sqrt{g^2 z(z-4r)}}{2g}$$

$$g > 0 \rightarrow \lambda_1 = \frac{-z - \sqrt{z(z-4r)}}{2} \quad \lambda_2 = \frac{-z + \sqrt{z(z-4r)}}{2}$$

Stability independent on d or g

Oscillations: possible if $\sqrt{z(z-4r)} < 0$
 $z^2 - 4rz < 0 \quad | : z$

If oscillations,
 equilibrium is stable:

$$\operatorname{Re}(\lambda_{1/2}) = -\frac{z}{2}$$

$$z - 4r < 0$$

$$z < 4r$$

$$\rightarrow$$

$$\frac{z}{r} < 4$$

z has to be more than 4-fold of r

if no oscillations:

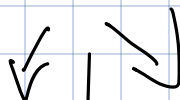
$$z(z-4r) > 0$$

$$\rightarrow z > 4r$$

$$\rightarrow \begin{matrix} z - 4r > 0 \\ z - 4r < z \end{matrix}$$

since $4r > 0$
 the term on the left can
 only become smaller

equilibrium is stable when



$$\frac{-z + \sqrt{z(z-4r)}}{2} < 0 \quad (\lambda_2)$$

$$(\lambda_2)$$

$$\frac{-z - \sqrt{z(z-4r)}}{2} < 0 \quad (\lambda_1)$$

$$(\lambda_1)$$

$$-z + \sqrt{z(z-4r)} < 0 \quad | +z|x^2$$

$$|+z|x^2$$

$$-z - \sqrt{z(z-4r)} < 0 \quad | +z|x^2$$

$$z^2 + 4zr < z^2 \quad | :z$$

$$z + 4r < z$$

$$z > 0 \text{ per definition} \rightarrow -z < 0$$

$$\sqrt{z(z-4r)} > 0 \quad \forall z \geq 4r \rightarrow -\sqrt{z^2 - 4zr} < 0$$

always negative!

→ Conclusion: equilibrium always stable, oscillation-free for $z \geq 4r$